NEMATODES

Root-knot nematode (*Meloidogyne incognita*): Infection by nematode causes swelling of roots known as root knot or galls, resulting in yellowing of foliage, defoliation, premature wilting and stunting of plants. Nematode-infected plants may have smaller leaves and fewer flowers.

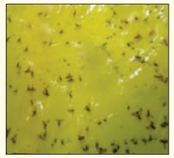
NUTRIENT DEFICIENCY / PHYSIOLOGICAL DISORDER

Boron/Sulphur deficiency: Boron/ potassium deficiency symptoms are common in bitter gourd/cucumber. Due to potassium deficiency, fruits fail to expand at stem end as in cucumber. Seed set and fruit development is affected by boron deficiency and fruit can taper and curve at the blossom end as in bitter gourd.

INTEGRATED PEST MANAGEMENT INTERVENTIONS FOR CUCURBITACEOUS VEGETABLE CROPS

- Seed treatment with *Trichoderma* @ 10 g/kg of seed and add @2.5 kg/acre soil along with well rotten FYM to manage wilt
- Two to three applications of neem 300 ppm @ 10 ml/litre against hadda beetle and red pumpkin beetle in early stages of crop growth.
- Need-based spray of DDVP 76 EC @ 500 g a.i./ha in the early stages for red pumpkin beetle control
- Two sprays of *Bacillus thuriengiensis var kurstaki** 2 WP @ 2g /litre for protection against cucumber moth on bitter gourd.
- Need based spray of chlorantraniliprole 18.5 SC @ 20-25 g a.i./ha for cucumber moth in bitter gourd.
- Installation of cue lure traps for fruit flies for wider area management @ 10/acre. Wooden plywood blocks are dipped in solution of Ethanol: Cuelure: Insecticide (DDVP) @ 8:2:1 for 48 hours.

- Raking of soil for exposing fruit fly pupae to sunlight and predatory fauna.
- Install yellow coloured sticky traps @ 10 /acre for fruit fly management.
- Collection and destruction of fruit fly infected fruits periodically.
- Prophylactic spray with *mancozeb 75 WP/zineb 75 WP @ 0.2 % against downy mildew, anthracnose and leaf spots.
- Use delta and blue coloured traps @ 5/acre for white fly and thrips, respectively. This may be followed by neem spray.
- Preferably grow cucumber on bamboo support or tresilising or in green houses can reduce downy mildew with reduced lead wetness and free air flow.
- Before any chemical pesticide spray, all mature fruits should be harvested.
- Need based application of neem cake @ 400 kg/ha for nematode management
- Need- based spray with cymoxanil 8% + mancozeb 64% @ 0.3% against downy mildew, carbendazim 50 WP @ 0.2 % and thiophenate methyl 70 WP @ 1430 g/1000 l water against powdery mildew and anthracnose, respectively.
- An application of 10 kg borax per hectare to deficient soil before planting will prevent boron deficiency. Foliar sprays of borax (100 g/100 L) may also be used



***No label claim but validated and found effective**

CONSERVATION OF NATURAL ENEMIES

Natural enemies of pests should be protected from unwanted & excessive sprays of chemical pesticides.

Tolerant / Moderately resistant Varieties / Hybrids of cucurbits

Crop	Pest	Tolerant/resistant variety/hybrid
Watermelon	Anthracnose, powdery mildew, downy mildew	**Arka Manik**
Musk Melon	Powdery mildew Downy mildew Cucumber green mottle mosaic virus	**Arka Rajhans** **Punjab Rasila** **DVRM-1**
Bottle gourd	Powdery mildew, downy mildew	**Narendra Rashmi**
Cucumber	Downy mildew and powdery mildew	**PCUCH 3**

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Integrated Pest Management Strategies for Cucurbits

भाकृअनुप
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INTRODUCTION

Cucurbitaceous vegetables are very important and large group of vegetables cultivated all over India extensively. These are very popular and highly remunerative vegetables grown mostly during summer and *Kharif* seasons and sometimes during winter in South and Western India. Cucurbits share about 5.6 % of total vegetable production in India with productivity around 10 t/ha which is quite low. One of the major constraints identified in the potential production system is the increasing incidence of insect pests, diseases and nematodes, often resulting in substantial yield losses and affecting the fruit quality. Due to their cultivation during different seasons/off seasons, economic losses due to various pests vary from 20-25 per cent. Cultivation of cucurbitaceous vegetables during different seasons of the year also results in continuous and abundant food supply for the buildup, perpetuation and multiplication of insect pests, diseases and nematodes.

In order to reduce the losses due to these pests, farmers are highly dependent on excessive use of chemical pesticides and it is not unusual for the vegetable growers to even give 10-15 sprays in these crops in a season without concomitant increase in productivity. In this scenario, the fruits of cucurbitaceous vegetables are likely to retain unavoidably high level of pesticide residues which are not only be hazardous to consumers but may affect the export quality as well. Further, the excessive dependence on chemicals leads to the problem of development of resistance, resurgence, and environmental pollution and health hazards to farmers and disturbs the delicate balance between pests and natural enemies. With a view to minimize all these problems and to create awareness among the farmers, integrated pest management strategies for major cucurbits *viz*; bitter gourd, bottle gourd and cucumber crops have been developed and validated at NCIPM, New Delhi and IIVR, Varanasi and are summarized as under:

INSECTS

Red pumpkin beetle (*Aulacophora foveicollis*): It is a major pest of bottle gourd and also attacks, pumpkin, cucumber and water melon as well. It feeds voraciously on leaves, flower buds and flowers. Damage may reach up to 75% at seedling stage. Grubs in soil, feed on roots and underground stem while adults feed on cotyledons and foliage causing death and retardation of growth. Grubs are creamy white while adults are bright orange red.

Leaf miner (*Liriomyza trifolii*): Adults are pale yellow while maggots are apodous, minute, and orange yellowish and pupate within the mines. Larvae make serpentine mines and feed within. Due to severe infestation, there is drying and dropping of leaves.

Cucumber moth (*Diphania indica*): Young larva scrapes the chlorophyll content and later on it folds and webs the leaves and feeds within. It also feeds on flowers and bores into developing fruits. Larvae are bright green with a pair of white mid-dorsal lines. Adults have whitish wings with broad and dark marginal patches. Eggs are laid singly or in small groups mostly on lower surface of leaves.

Melon fruit fly (*Bactrocera cucurbitae*): It is widely distributed, most active from July to September and damages all cucurbits particularly bitter gourd and musk melon.

Gravid female inserts the eggs 2 to 4 mm deep in the soft, tender, fruit tissues, and the maggots feed inside the fruit tissues and make galleries. Fruits subsequently rot or become distorted. Punctures serve as entry point for bacteria and fulgi fly. Pupation occurs in the soil at 0.5 to 15 cm below the soil surface.

Hadda beetle (*Epilachna dodecastigma, Henosepilachna vigintioctopunctata*): It is minor pest and is active from August to October. Both the grubs and adults feed on the upper surface of leaves. The leaves are eaten between the veins, sometimes being completely stripped to the mid-rib. The leaves present a lace-like appearance; turn brown, dries up and falloff. Thus, the plant is completely skeletonized.

White fly (*Bemisia tabaci*): It is a major pest of bitter gourd and bottle gourd. Both the nymphs and adults suck the plant sap mainly from under-side of the leaves and secrete honeydew on which black sooty-moulds develops which later in turn reduces photo-synthesis by the plants. Apart from causing direct damage by their feeding, they also act as vector of Gemini virus

Mirid bug (*Nesideocoris* sp): Adults and nymphs pierce the tender leaves and shoots, growing flower buds and tender fruits with their stylets and suck sap on bottle gourd. Because of feeding, damaged fruits exhibit punctures and reddish brown stains from which sap oozes out. This reduces fruit quality that fetches lower market value.

DISEASES

Root rot caused by *Rhizoctonia solani* have slightly sunken lesions at or below soil level with above ground symptom include low vigor, stunting and wilting of seedlings.

Downy mildew (*Pseudoperonospora cubensis*): It is one of the serious diseases which initiates as water soaked angular lesions under conditions of high humidity and moderate temperature, soon turning chlorotic and finally on the corresponding lower surface purplish downy growth is seen.

Powdery mildew (*Sphaerorheca fuliginea*): The characteristic white, powdery fungal growth is readily recognizable on leaves, stems and vines. It is observed on almost all cucurbits and causes considerable loss. Affected leaves wither and dry. Powdery mildew affects fruit quality and reduces yield by reducing number and size of fruit.

Anthracnose (*Colletotrichum lagenarium*): Yellowish water soaked spots appear which enlarge and turn dry brownish black with centre giving shot hole appearance. On fruits spots are circular sunken with dark borders containing numerous pin head size fruiting bodies.

Cercospora leaf spot (*Cercospora citrullina*): The disease occurs on almost all cucurbits. On leaves, lesion develop as small grey or white yellow regular or irregular spots with yellow halo. These spots later enlarge and eventually encircle the leaf. Centre of lesions may become brittle and crack. Symptoms with white centre and brown margin spots also observed in petiole and main stem.

***Fusarium* wilt (*Fusarium oxysporum*):** Yellowing, drooping of leaves and wilting and drying of whole plant are the major symptoms. Vascular bundles in the collar/root region become discoloured/brown.

Virus: This begomo virus transmitted by whitefly is a severe constraint in cucurbits which causes leaf curl, yellow mosaic and stunting. In cucumber mosaic, typical mosaic symptoms develop on young leaves which mottle with alternating light green and dark green patches.